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320452(20)

B. E. (Fourth Semester) Examination, April-May 2016

(New Scheme)

(Civil Engg. Branch)

FLUID MECHANICS-II

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *Attempt all questions. Part (a) from each question is compulsory. Attempt any two parts from parts (b), (c) and (d) of each question.*

Unit-I

1. (a) What is turbulence? Give examples of common occurrence which characterize turbulent flow. 2

(b) Derive the Karman-Prandtl velocity equations and

discuss the role of laminar sub-layer and surface roughness on velocity distribution. 7

(c) A pipe of dia 20 cm and length 2000 m connects two reservoirs, having difference of water levels as 20 m. Determine the discharge through the pipe. If an additional pipe of dia. 20 cm and length 1200 m is attached to the last 1200 m length of the existing pipe, find the increase in the discharge. Take $f = 0{\cdot}015$ and neglect minor losses. 7

(d) Short notes on : 7

(i) Colebrook-White equation

(ii) Moody's diagram

(iii) Hardy-cross method

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Unit-II

(a) Define Laminar boundary layer and turbulent boundary layer. 2

(b) Derive an expression for the energy thickness. 7

(c) For the velocity profile for laminar boundary layer

$\frac{u}{U} = \frac{3}{2}(y/\delta) - \frac{1}{2}(y/\delta)^3$. Determine the boundary layer thickness, shear stress, drag force and coefficient of drag in terms of Reynold no. (Re). 7

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(d) A Jet Plane which weights 29·43 kN and having a wing area of 20 m^2 flies at a velocity of 950 km/hour. When the engine delivers 7357·5 kW power 65% of the power is used to overcome the drag resistance of the wing. Calculate the co-efficients of lift and drag for the wing. The density of the atmospheric air is 1·21 kg/m^3. 7

Unit-III

3. (a) Explain the terms : specific energy and specific energy curve. 2

(b) Explain the terms of hydraulic jump. Derive an expression for the depth of hydraulic jump. 7

(c) Show that in a rectangular channel : 7

(i) Critical depth is two-third of specific energy, and

(ii) Froude number at critical depth is unity

(d) Find the slope of the free water surface in a rectangular channel of width 20 m having depth of flow 5 m. The discharge through the channel is 50 m^3/s. The bed of the channel is having a slope of 1 in 4000. Take the value of chezy's constant 'C' = 60. 7

Unit-IV

4. (a) What is dimensional analysis? What are the uses of dimensional analysis? 2

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(b) What are the different laws on which models are designed for dynamics similarity? Where are they used? 7

(c) The pressure difference ΔP in a pipe of diameter D and length l due to viscous flow depends on the velocity V, viscosity μ and density ρ. Using Buckingham's π-theorem, obtain an expression for ΔP. 7

(d) The water is flowing with a velocity of 1·5 m/s in a pipe of length 2500 m and of diameter 500 mm. At the end of the pipe, a valve is provided. The value is closed in 2 sec. find the rise in pressure behind the valve. Assume the pipe to be rigid one and take Bulk modulus of water. $K = 19{\cdot}62 \times 10^4\ N/cm^2$. 7

Unit-V

5. (a) Differentiate between the turbine and pumps. 2

(b) Obtain an expression for unit speed, unit discharge and unit power for a turbine. 7

(c) A pelton wheel is to be designed for a head of 60 m when running at 200 r.p.m. The pelton wheel develops 95·6475 kW shaft power. The velocity of buckets = 0·45 times the velocity of the Jet, overall efficiency = 0·85 and co-efficient of the velocity is equal to 0·98. 7

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(d) The dia. of a centrifugal pump, which is discharging 0·03 m^2/s of water against a total head of 20 m is 0·40 m. The pump is running at 1500 r.p.m find the head, discharge and ratio of power of a geometrically similar pump of dia 0·25 metre when it is running at 3000 r.p.m. 7

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